Basic Concepts of Navigation

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Basic Concepts of Navigation
Part I

- Overview
- The Navigation Problem and Reference Frames
- Inertial Navigation

Overview

- Etymology of the term 'Navigation': Latin *navis*='ship' and *agere*='to move or direct'
- Arose out of marine or ship navigation due to the challenge of navigating without landmarks
- Modern distinction between navigation and guidance
  - navigation associated with state determination (open loop)
  - guidance associated with control feedback (closed loop)

The Navigation Problem from a Mathematical Perspective

With respect to some mathematical reference frame, determine ...

- vector displacement (3d position) and its time derivatives (velocity, acceleration, ...)
- rotational orientation (3d attitude) and its time derivatives (angular velocity, angular acceleration, ...)
- Note: This is a pure mathematics problem ... sensors do not enter into it

The Navigation Problem from a Mathematical Perspective

- Reference frame is crucial:
  - Position: origin and three linearly independent vectors yields three numbers
  - Attitude: rotations about three linearly indepedent vectors yields three numbers
  - I.e., 6 DOF = six degrees of freedom (3 position coordinates + 3 attitude coordinates)

The Navigation Problem from a Real World Physical Perspective

- Which reference frame should we use?
- Rectilinear or curvilenar coordinate system?

The Navigation Problem from a Real World Physical Perspective

Position

For terrestrial navigation, global spherical (spheroidal) coordinates are used:

- Latitude: ± 90 degrees north(+) or south(-) of the equator.
- Longitude: ± 180 degrees east(+) or west(-) of the prime meridian through Greenwich, England
- Height: Above the ellipsoid (normal to surface), or radius from center of earth

The Navigation Problem from a Real World Physical Perspective

Direction: Bearing, Course, Track

- North Reference: True and Magnetic. Direction angles are usually referenced clockwise with respect to north.
- Bearing: The angle with respect to north from one point on earth to another
- Track: The angle made by velocity vector with respect to north
- Course: Sometimes used synonym for track; sometimes expresses desired track vs. actual track
- Heading: Direction vehicle is pointing in with respect to north

The Navigation Problem from a Real World Physical Perspective

The Need for Sensors

- Reference frames are necessary, but not sufficient for real world navigation
- Also need real world sensor measurement data indicative of locations, distances, angles, etc. with respect to these reference frames
- But real world sensors are not perfect! They only provide apparent positions, distances, angles, etc.

The Navigation Problem and Reference Frames

Topics

- Basic Navigation Problem and Solutions
- Earth Referenced Navigation
- Coordinate Frame Conversion
- Basic Strapdown Navigation Equations

Dead Reckoning

- Process of determining new position based on extrapolating from previously known position and velocity.
- Velocity = Speed and Direction
- Unaided inertial navigator is essentially a dead reckoning computer

Earth Referenced Navigation

- Terrestial navigation seeks to answer nav questions with respect to earth
  - Global earth reference
  - Local or relative navigation
- Complications:
  - Earth surface is curved (ellipsoid)
  - Earth is turning (can usually ignore motion about the Sun for terrestrial nav)

Earth Model: WGS 84

- Earth is not spherical, but oblate spheroid or ellipsoid
- Flattened at the poles
- World Geodetic System 84:
  - Semi-major axis: 6,378,137.0 m
  - Semi-minor axis: 6,356,752.314 m
- Not the same as geoid: no terrain or gravity model

Coordinate Frames

- Inertial
- Earth Fixed
- Local Level
- Body

Coordinate Frames

- Technically speaking, all frames are earth-centered
  - Common origen needed for linear transformations (such as rotations)
  - Can compensate by using a translation
- Practically speaking, we think of the Body frame as centered in the body of the vehicle (usually at center of gravity).
- Local Level is thought to be located at surface of earth directly beneath the vehicle.

Inertial Reference Frames

- Non-Rotating
- Non-Accelerating
- Fixed or moving uniformly with respect to another inertial frame (fixed stars)

Inertial Reference Frames

- Nav problem "easy" in inertial frames
- Integrate gyro rates once to get attitude (rotational motion independent of lateral motion)
- Rotate accelerometer outputs to inertial frame using attitude matrix, then integrate to get inertial velocity; integrate velocity to get position.
- Would still have to contend with instrument imperfections.

Inertial Reference Frame Integrations

$$\ddot{r}(t) = a_{\text{sf}} + \text{other external forces} \tag{1}$$

$$v(T) = \dot{r}(T) = v(0) + \int_0^T a_{\text{sf}} dt + \int_0^T (\text{other external forces}) dt \tag{2}$$

$$r(T) = r(0) + \int_0^T v(t) dt \tag{3}$$

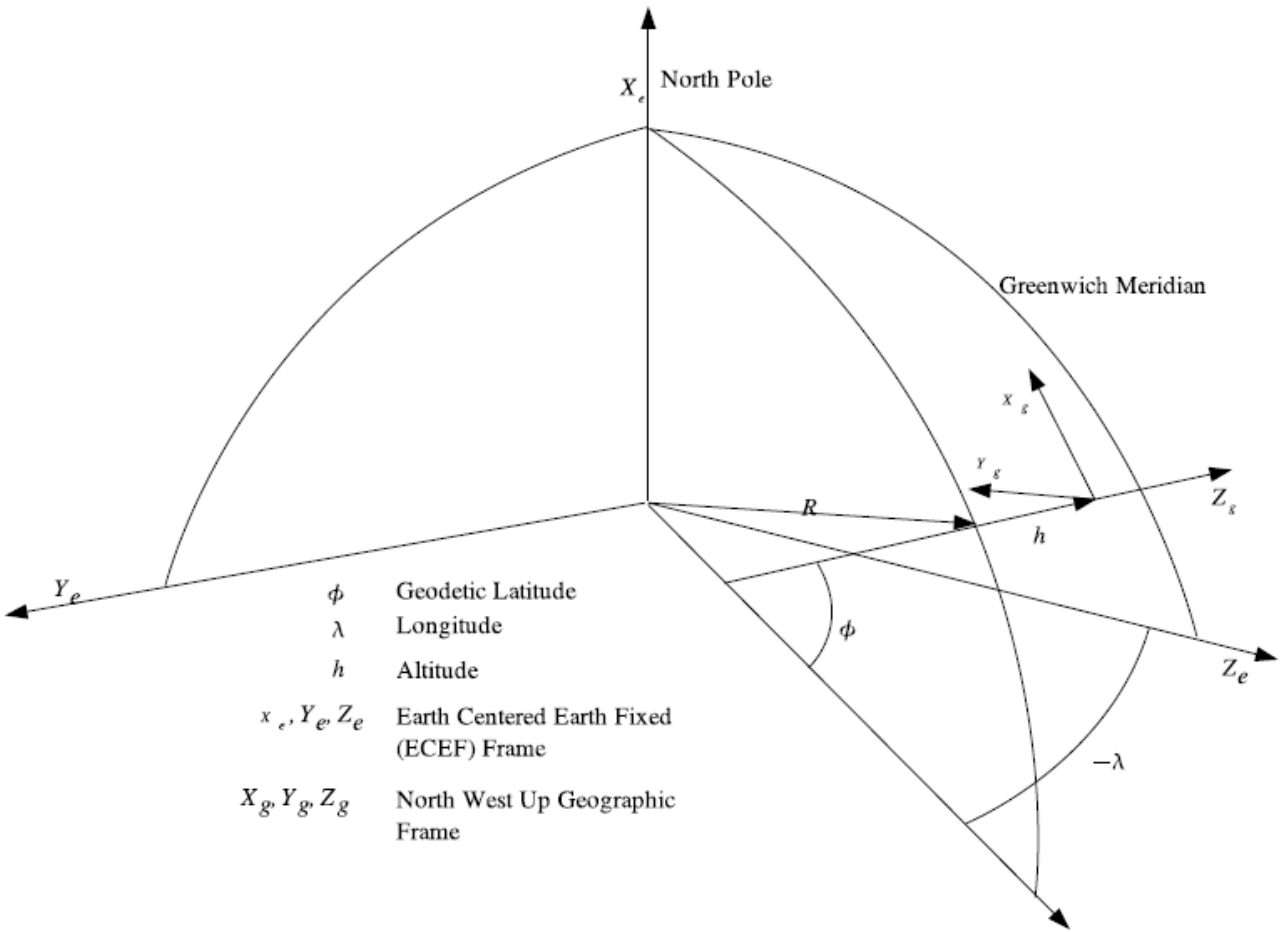
Earth Fixed = ECEF = Earth Centered Earth Fixed

- Rotating frame, i.e., non-inertial
- Cartesian (X, Y, Z in meters)
- Spherical (latitude ϕ, longitude λ, altitude
  - geodetic

Local Level

- Tangent to earth's surface but translated to center of earth.
- Vertical Z-axis is normal to earth's surface
- X, Y-axes are level
- Several kinds: Y north-slaved, wander (0 transport rate) or free azimuth
- Y makes wander angle α with respect to true north (ENU=East, North, Up).

Earth Centered, Earth Fixed: ECEF



Body Frame

- Attached to vehicle
- For aircraft:
  - X out the nose
  - Y out the right wing
  - Z down

Body Frame (cont)

- For space vehicles RIC:
  - R downward on radius to center of earth
  - I in-track, in direction of motion
  - C cross-track, perpendicular to motion
- Also sometimes referred as RAC, where 'A' stands for 'along-track'

Coordinate Frame Conversion

- All frames earth-centered, differing only by rotation
- Rotation represented by direction cosine matrix (DCM) or set of four quaternions
  - Expresses angles made by basis vectors of one frame with those of another
- C_A^B = DCM transformation from frame A to frame B

Coordinate Frame Conversion

- "Direction Cosine" comes from dot product:

$$\langle x, y \rangle = x \cdot y = \|x\| \, \|y\| \cos \theta_{xy}$$

  where θ_{xy} is the angle between the two vectors x and y.
- If x and y are unit vectors, then $\|x\| = \|y\| = 1$ and we have:

$$\langle x, y \rangle = x \cdot y = \cos \theta_{xy}$$

Coordinate Frame Conversion

- If $U = \{u_1, u_2, u_3\}$ and $V = \{v_1, v_2, v_3\}$ are two sets of orthonormal basis vectors, then the DCM relating one frame to the other is given by:

$$C_U^V = \begin{bmatrix} \langle u_1, v_1 \rangle & \langle u_1, v_2 \rangle & \langle u_1, v_3 \rangle \\ \langle u_2, v_1 \rangle & \langle u_2, v_1 \rangle & \langle u_2, v_3 \rangle \\ \langle u_3, v_1 \rangle & \langle u_3, v_2 \rangle & \langle u_3, v_3 \rangle \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta_{11} & \cos\theta_{12} & \cos\theta_{13} \\ \cos\theta_{21} & \cos\theta_{22} & \cos\theta_{23} \\ \cos\theta_{31} & \cos\theta_{32} & \cos\theta_{33} \end{bmatrix}$$

Coordinate Frame Conversion

- If $V = \{v_1, v_2, v_3\}$ is considered to be the reference frame and its units vectors are the standard basis so that

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

  then the DCM simplifies to just the coordinates of u_1, u_2, u_3 vectors with respect to V:

$$C_U^V = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix}$$

Coordinate Frame Conversion

- DCMs are (real) orthogonal matrices
  $\Rightarrow \left[C_A^B\right]^{-1} = \left[C_A^B\right]^\top = C_B^A$
- Third row (or column) is vector cross product of other two
  $\Rightarrow$ only six independent parameters
- DCMs satisfy: $C_B^A C_C^B = C_C^A$

Coordinate Frame Conversion

- Orthogonal matrices preserve lengths of vectors: $\|Rx\| = \|x\|$
- Orthogonal matrices also preserve angles between vectors: $\langle Rx, Ry \rangle = \left\langle R^\top Rx, y \right\rangle = \langle Ix, y \rangle = \langle x, y \rangle$
- Intuitively obvious in the case of rotations

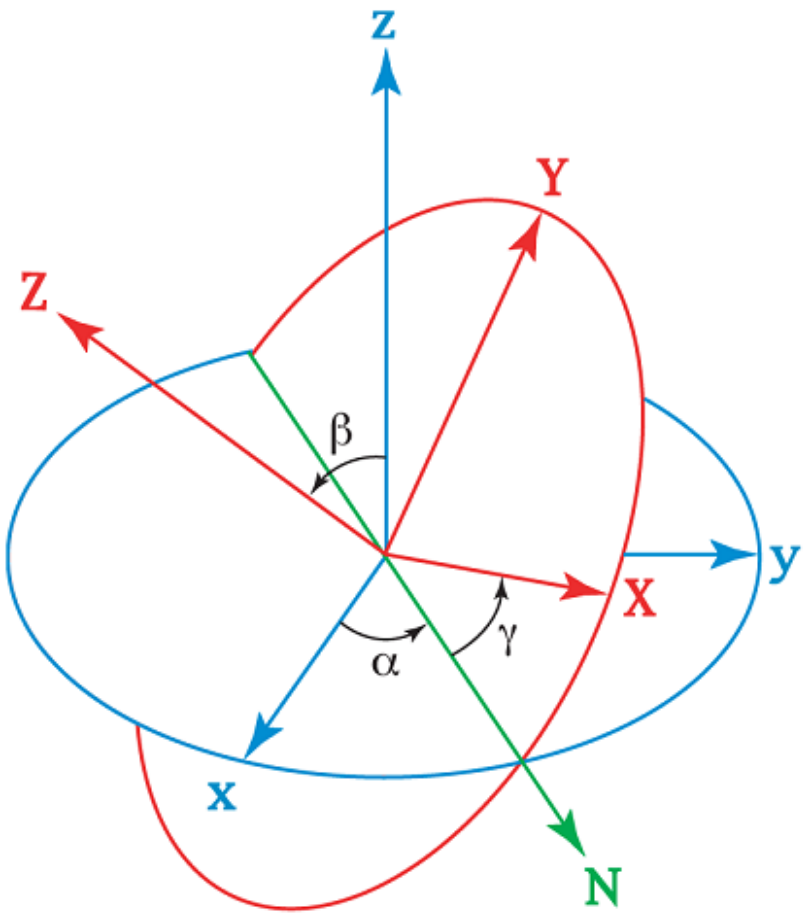
DCMs and Euler Rotation

- Euler showed that three simultaneous rotations about three orthogonal axes can be viewed as a single rotation about a single axis
- The single angle θ of rotation for a DCM can be determined from the eigenvalues of the DCM, which are:

$$\{1, \exp(i\theta), \exp(-i\theta)\}$$

- The determinant of a DCM is therefore always 1. Why? (Hint: How do you compute the determinant from the eigenvalues?)

Euler Rotation



DCMs and Euler Rotation

- The axis of rotation for the single Euler rotation can be found from DCM as follows:

$$C_U^V = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{bmatrix}$$

then the axis of rotation $r = [r_1, r_2, r_3]^\top$ is given by

$$\begin{aligned} r_1 &= u_{12}u_{23} - (u_{22} - 1)u_{13} \\ r_2 &= u_{21}u_{13} - (u_{11} - 1)u_{23} \\ r_3 &= (u_{11} - 1)(u_{22} - 1) - u_{12}u_{21} \end{aligned}$$

DCMs and Quaternions

- If C is a direction cosine matrix with elements c_{ij}, then the following four quantities define a quaternion of the rotation:

$$q_1 = \frac{1}{4q_4}(c_{23} - c_{32})$$
$$q_2 = \frac{1}{4q_4}(c_{13} - c_{31})$$
$$q_3 = \frac{1}{4q_4}(c_{21} - c_{12})$$
$$q_4 = \pm\frac{1}{2}\sqrt{1 + c_{11} + c_{22} + c_{33}}$$

DCMs and Quaternions

- It's even simpler to obtain quaternions from Euler rotation angle θ and axis given by a unit vector u:

$$q_i = u_i \sin(\theta/2), \text{ for } i = 1, 2, 3$$

and

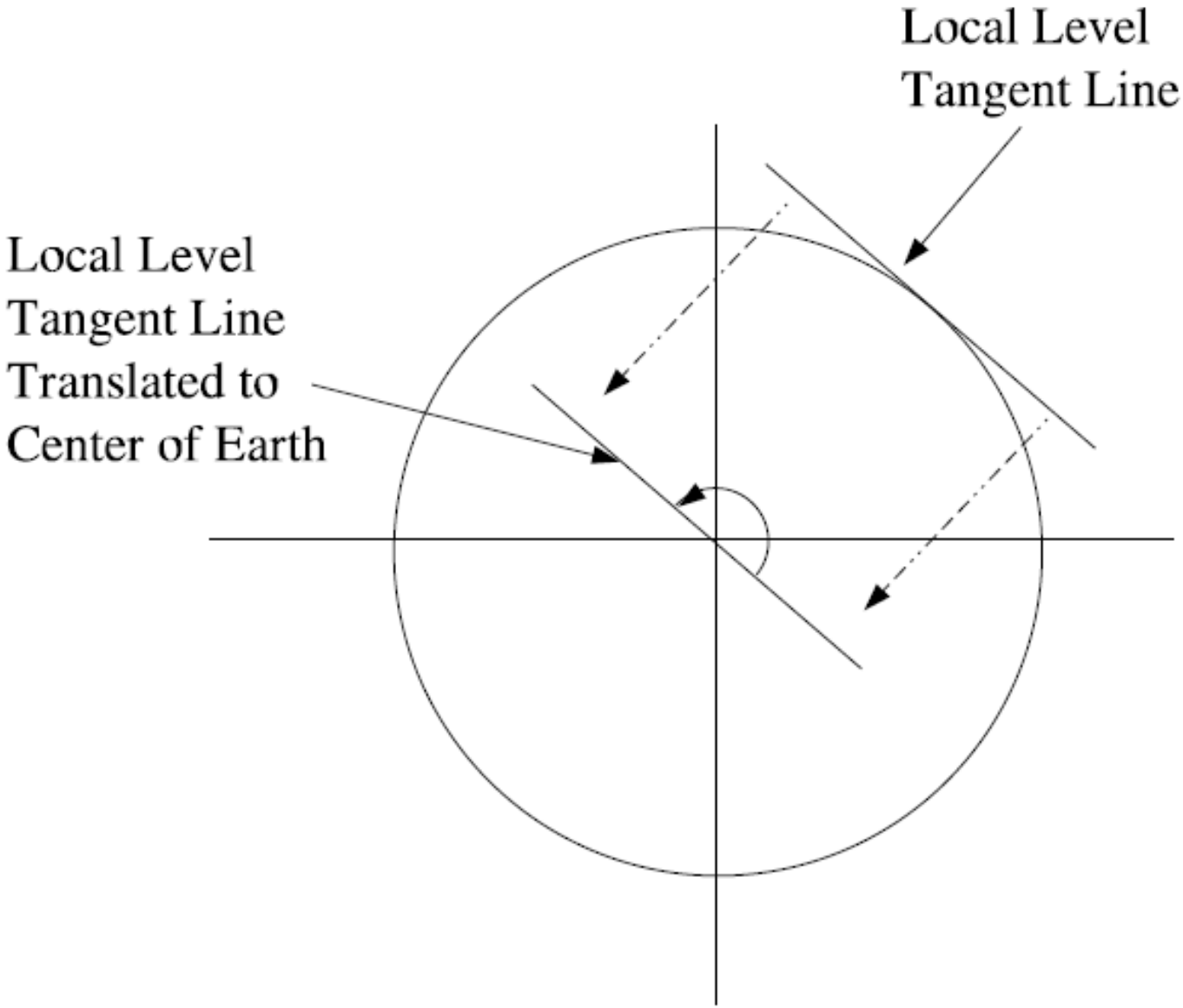
$$q_4 = \cos(\theta/2)$$

Coordinate Frame Conversion: Most important DCMs for terrestrial navigation

- Body to Local Level: C_B^L
  - sometimes called Body to Nav and notated C_B^N
  - yields attitude info (Euler angles for roll, pitch, yaw)
- Local Level to Earth: C_L^E
  - yields position info and also used to transform earth rate to local level

Coordinate Frame Conversion: Most important DCMs

- Visualize C_L^E's determination of position as follows:
  - As vehicle moves, local level frame changes its orienation with respect to earth's center in order to maintain local level with respect to earth's surface
  - If you think of local level frame as attached to earth's surface, you get distracted by the displacement of the frame
  - If you move local level frame to center of earth, the displacement disappears and all that remains are the rotations


Local Level
Tangent Line
Local Level
Tangent Line
Translated to
Center of Earth

- Gimbaled INS:
  - Gyro stabilized platform
  - Gimbal assembly and torque motors
  - Resolvers, slip rings, bearings, etc. to control platform
  - Goal: Maintain stable, locally level or inertial reference frame

Strapdown Navigation: Strapdown vs. Gimbaled

- Gimbaled INS Issues
  - Reliability and Cost: Lots of moving parts!
  - Body rate and acceleration information not directly available
  - Rate signals are generally noisy and have low bandwidth

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS:
  - Inertial instruments strapped to body frame
  - Requires local level accelerations to be computed using direction cosine matrix C_B^L
  - C_B^L derived from gyro-sensed body rates
  - Strapdown systems can be implemented with either rate gyros or attitude gyros

Strapdown Navigation: Strapdown vs. Gimbaled

- Strapdown INS Advantages
  - Immediate detection of body rates and accelerations
  - Fewer moving parts
  - Potential redundancy advantages
- Gimbaled systems are nearly obsolete

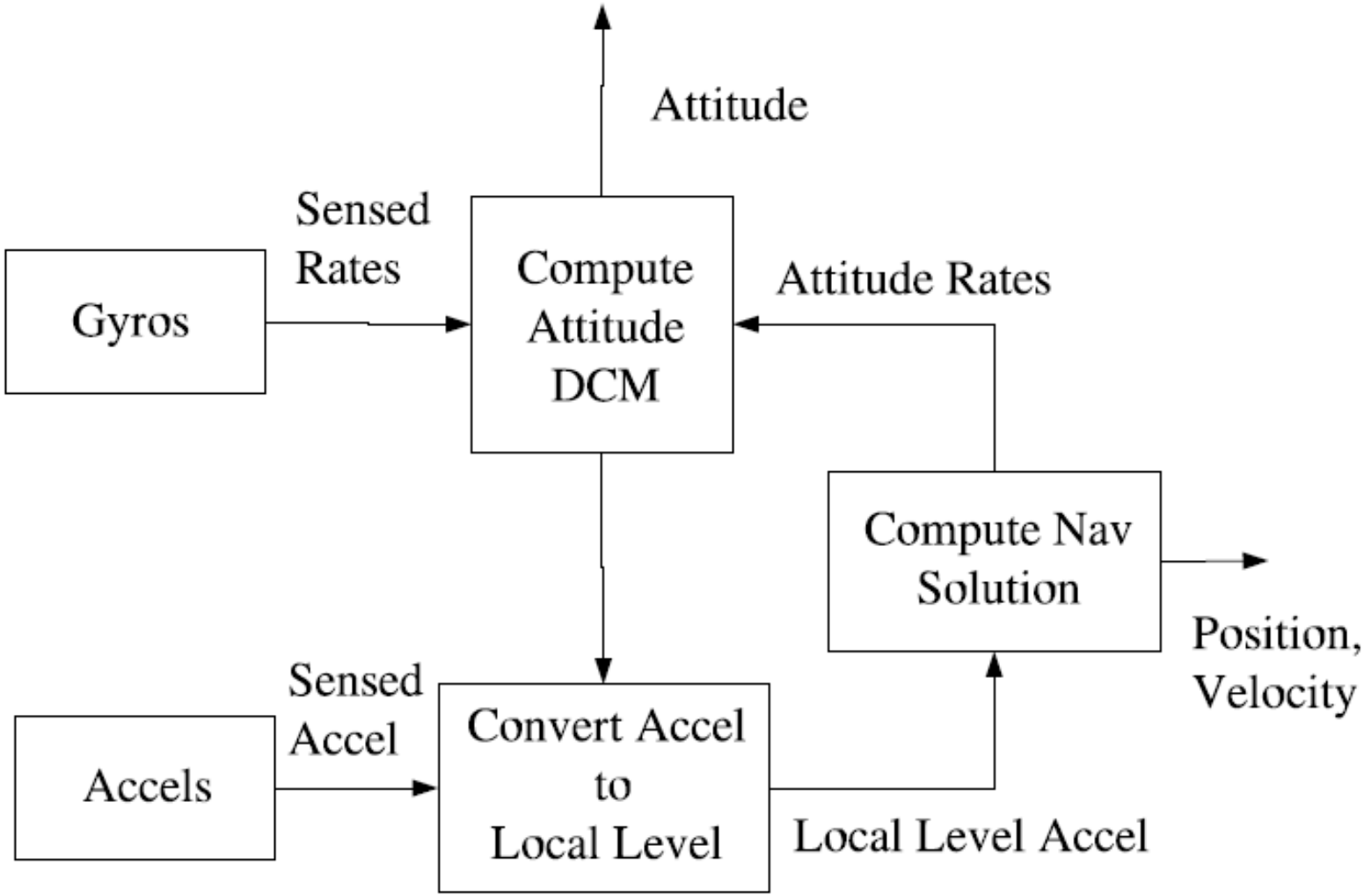


Figure: Strapdown System Block Diagram

Basic Strapdown Navigation Equations

- Need to compute attitude, velocity, and position
- I.e., need to compute C_B^L and C_L^E
- These DCMs are dynamic quantities
- C_B^L determines attitude and can be rapidly varying
- C_L^E determines position and is slowly varying (relative to attitude dynamics)

Strapdown Navigation Equations

- Three differential equations
  - Equation for body to local level transformation C_B^L
  - Equation for velocity (uses C_B^L)
  - Equation for local level to earth for position DCM C_L^E

Strapdown Navigation Equations

- Two Algebraic Equations
  - Equation for converting velocity to transport rate (to be explained)
  - Equation for converting earth rate to local level coordinates

Strapdown Navigation Equations

The Three Differential Equations

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L \tag{4}$$

$$\dot{v}^L = C_B^L a_{\mathsf{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L \tag{5}$$

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L \tag{6}$$

$$\omega_{EL}^{L} = \frac{1}{r} u_r^L \times v^L \text{ (wander azimuth } \rho_z = 0) \tag{7}$$

$$\omega_{IE}^{L} = \left[C_L^E \right]^\top \omega_{IE}^{E} \tag{8}$$

Strapdown Navigation Equations

The C_B^L Equation (4)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- All the Ω matrices represent rotation vector cross products in skew-symmetric matrix format
  - Ω_{IB}^B is rotational rate of aircraft in body coordinates
  - Ω_{IE}^L is earth rate in local level coordinates
  - Ω_{EL}^L is transport rate in local level coordinates

Strapdown Navigation Equations: Detour into Cross Product Matrices

- The cross product of $\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$ with any other vector v is the same as multiplying v on the left by $\Omega = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$
- $\omega \times v = \Omega v$

Strapdown Navigation Equations

The C_B^L Equation (4)

$$\dot{C}_B^L = C_B^L \Omega_{IB}^B - \left(\Omega_{IE}^L + \Omega_{EL}^L\right) C_B^L$$

- Transport rate is the rotational rate of the aircraft due to motion over the surface of the earth. Is a function of velocity.
- Ω_{IB}^B is total inertial angular rate sensed by gyros and includes effects of transport and earth rates.
- Hence transport and earth rate effects have to be subtracted off.

Strapdown Navigation Equations

The Strapdown Velocity Equation

(5)

$$\dot{v}^L = C_B^L a_{\text{sf}}^B + g' - \left(2\Omega_{IE}^L + \Omega_{EL}^L\right) v^L$$

- Sometimes referred to as the velocity rate equation
- a_{sf}^B is specific force sensed by accelerometers in body coordinates (because the accels are strapped to the body – hence the term strapdown)

Strapdown Navigation Equations

The Transport Rate Equation

(6)

$$\dot{C}_L^E = C_L^E \Omega_{EL}^L$$

- Connected to velocity rate equation by the first algebraic equation (7):

$$\omega_{EL}^L = \frac{1}{r} u_r^L \times v^L$$

- Eq. (6) Sometimes referred to as position rate equation, because C_L^E directly implies the vehicle position

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Geographic)

- North transport rate:

$$\rho_N = \frac{v_E}{R_e}\left[1 - \frac{h}{R_e} - e\sin^2\phi\right] \tag{9}$$

- East transport rate:

$$\rho_E = -\frac{v_N}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3\cos^2\phi\right)\right] \tag{10}$$

where e is ellipticity (approx. $\frac{1}{298}$) and R_e is mean equatorial earth radius

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Geographic)

- Latitude Rate:

$$\dot{\phi} = -\rho_E \tag{11}$$

- Longitude Rate:

$$\dot{\lambda} = \rho_N \sec \phi \tag{12}$$

Strapdown Navigation Equations

- Geographic (north-slaved) mechanization has singularities at poles
- Two other approaches to local level coordinates:
  - Free azimuth: vertical component of transport rate $\rho_z =$ additive inverse of vertical component of earth rate
  - Wander Azimuth: vertical component of transport rate $\rho_z = 0$

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- X Transport Rate:

$$\rho_x = -\frac{v_y}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{22}^2 - d_{21}^2\right)\right] - \frac{v_x}{R_e}\left(2d_{21}d_{22}\right)e \tag{13}$$

- Y Transport Rate:

$$\rho_y = \frac{v_x}{R_e}\left[1 - \frac{h}{R_e} - e\left(1 - 3d_{21}^2 - d_{22}^2\right)\right] + \frac{v_y}{R_e}\left(2d_{21}d_{22}\right)e \tag{14}$$

Strapdown Navigation Equations

Position Rate / Transport Rate Equations (Wander and Free Azimuth)

- Azimuth Rate:

$$\rho_z = \begin{cases} -\Omega_e d_{23}, & \text{if free azimuth;} \\ 0, & \text{if wander azimuth} \end{cases}$$

- In equations above, the matrix $[d_{ij}]$ is the direction cosine matrix C_L^E from local level (L) frame to earth centered earth fixed (E) frame.

GPS

- Trilateration: Knowing the distance from an unknown point to three known points yields the coordinates of the unknown point
- What if the distances are not known precisely? Answer: Use 'pseudorange' instead, i.e., use signals traveling at speed of light and very accurate clocks
- What if the coordinates of the three points are not really known precisely? Answer: Use your best guess!

Trilateration

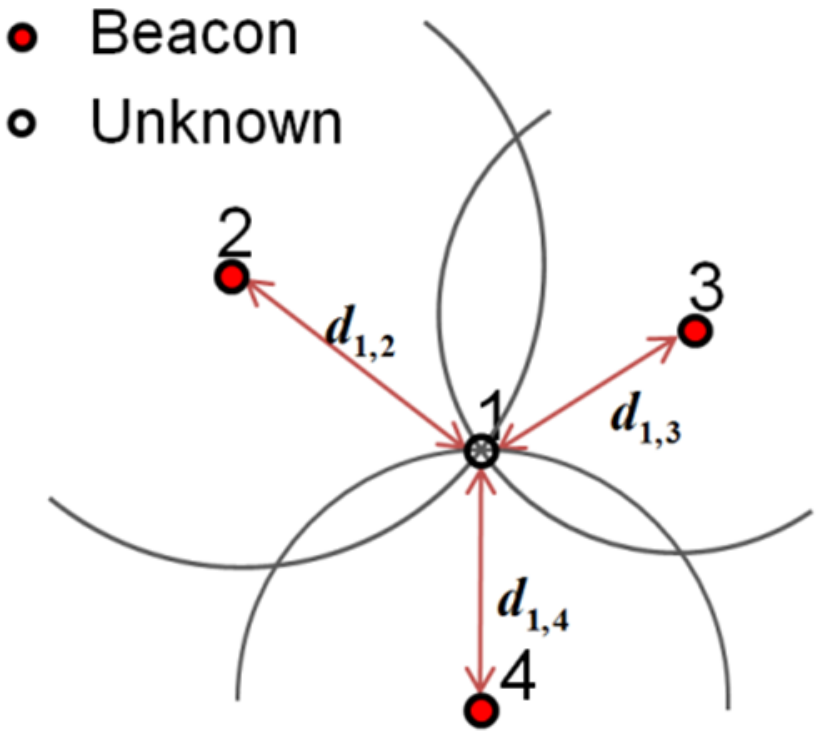


Figure: Trilateration: Distances to Three Beacons

Pseudorange

- distance = (speed of signal) $\times$ (time of signal flight)
  - $\rho = c\Delta t$
- Called 'pseudorange' because it is not a true range; it is only apparent due to inaccurate clocks
- By the same token, the coordinates of the transmitting satellites are also not known precisely, and so we could speak of 'pseudopositions' as well. But nobody does that!

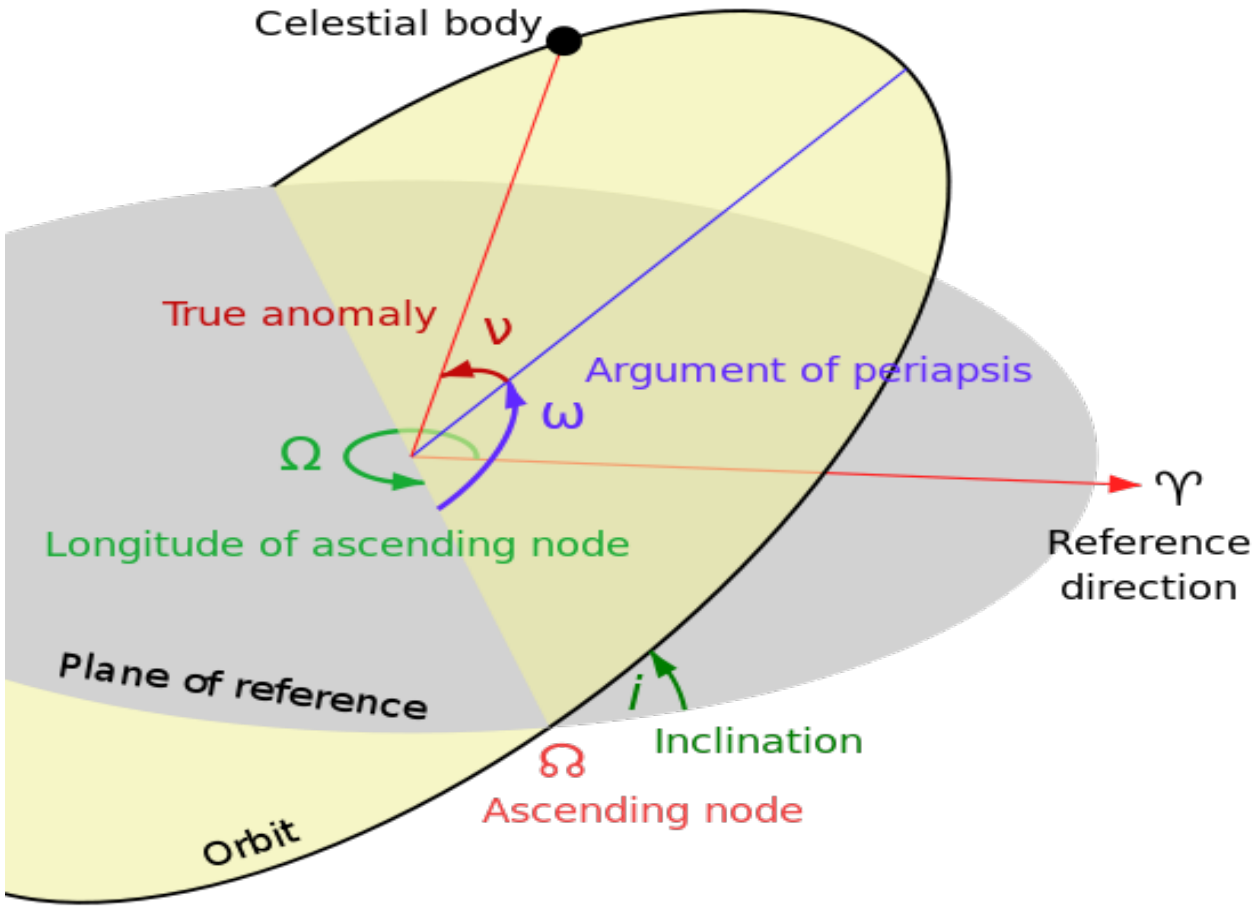
Satellite Position

- How do we know which satellite is transmitting? Answer: Unique Pseudorandom Noise (PRN)
- How we know the position of the satellite at time of transmission? Answer:
  - Six Keplerian elements
  - User receiver converts to WGS-84 X,Y,Z ECEF coordinates

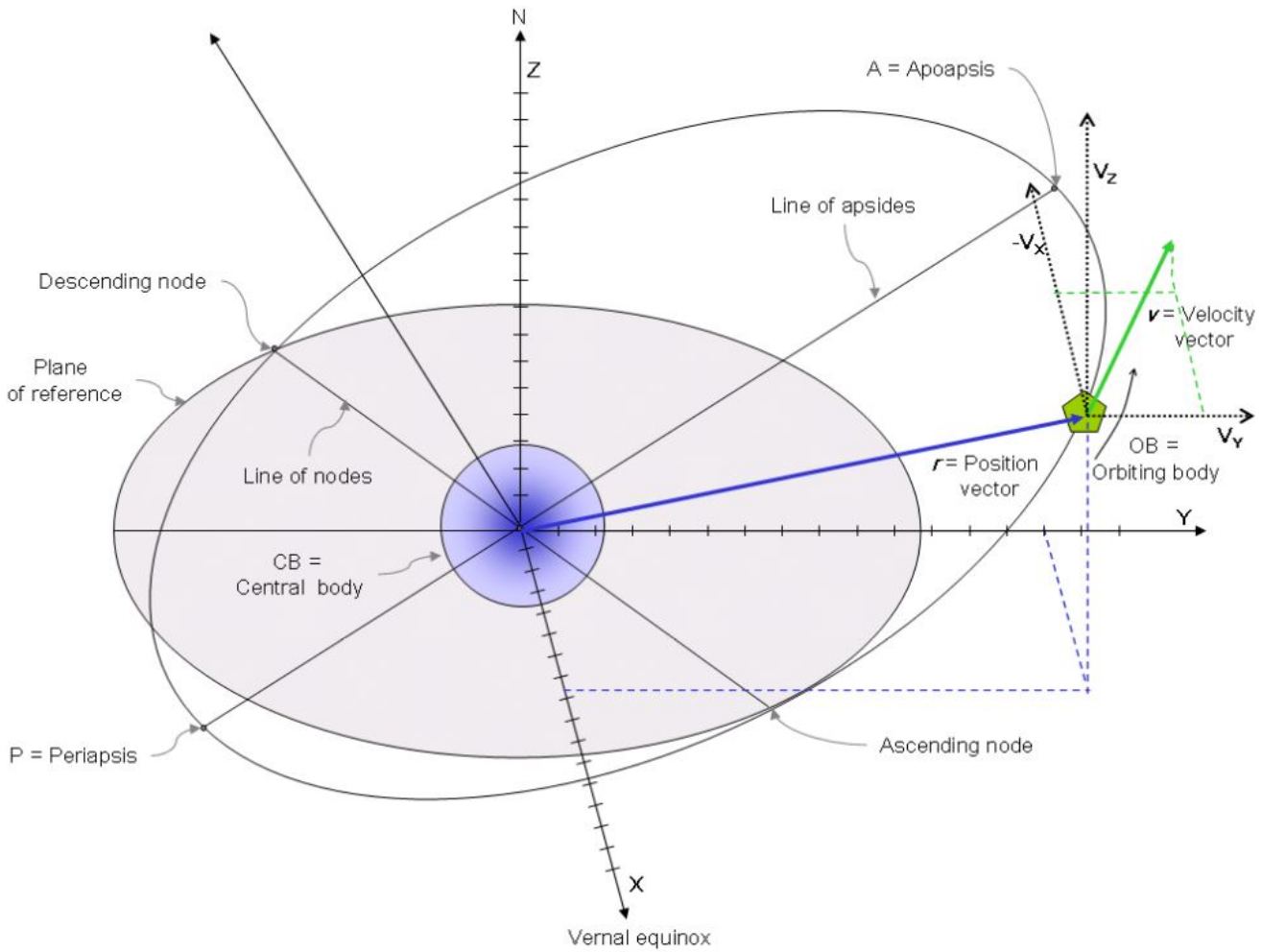
Six Keplerian Elements

- Eccentricity
- Semi-major axis
- Inclination
- Longitude of the ascending node
- Argument of periaspsis
- Mean anomaly

Keplerian Elements



Keplerian Elements



IS-GPS-200 Table 20-II

1 of 2

Table 20-II. Ephemeris Data Definitions	
M_0	Mean Anomaly at Reference Time
Δn	Mean Motion Difference From Computed Value
e	Eccentricity
$\sqrt{A}$	Square Root of the Semi-Major Axis
Ω_0	Longitude of Ascending Node of Orbit Plane at Weekly Epoch
i_0	Inclination Angle at Reference Time
ω	Argument of Perigee
$\dot{\Omega}$	Rate of Right Ascension
IDOT	Rate of Inclination Angle

IS-GPS-200 Table 20-II

2 of 2

C_{uc}	Amplitude of the Cosine Harmonic Correction Term to the Argument of Latitude
C_{us}	Amplitude of the Sine Harmonic Correction Term to the Argument of Latitude
C_{rc}	Amplitude of the Cosine Harmonic Correction Term to the Orbit Radius
C_{rs}	Amplitude of the Sine Harmonic Correction Term to the Orbit Radius
C_{ic}	Amplitude of the Cosine Harmonic Correction Term to the Angle of Inclination
C_{is}	Amplitude of the Sine Harmonic Correction Term to the Angle of Inclination
t_{oe}	Reference Time Ephemeris (reference paragraph 20.3.4.5)
IODE	Issue of Data (Ephemeris)

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (1 of 3)

Table 20-IV. Elements of Coordinate Systems (sheet 1 of 2)	
$\mu = 3.986005 \times 10^{14}$ meters^3/sec^2	WGS 84 value of the earth's gravitational constant for GPS user
$\dot{\Omega}_e = 7.2921151467 \times 10^{-5}$ rad/sec	WGS 84 value of the earth's rotation rate
$A = (\sqrt{A})^2$	Semi-major axis
$n_0 = \sqrt{\frac{\mu}{A^3}}$	Computed mean motion (rad/sec)
$t_k = t - t_{oe}$*	Time from ephemeris reference epoch
$n = n_0 + \Delta n$	Corrected mean motion

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (2 of 3)

$M_k = M_0 + nt_k$	Mean anomaly
$M_k = E_k - e \sin E_k$	Kepler's Equation for Eccentric Anomaly (may be solved by iteration) (radians)
$\nu_k = \tan^{-1}\left\{\frac{\sin \nu_k}{\cos \nu_k}\right\}$ $= \tan^{-1}\left\{\frac{\sqrt{1-e^2}\sin E_k / (1 - e\cos E_k)}{(\cos E_k - e)/(1 - e\cos E_k)}\right\}$	True Anomaly

* t is GPS system time at time of transmission, i.e., GPS time corrected for transit time (range/speed of light). Furthermore, t_k shall be the actual total time difference between the time t and the epoch time t_{oe}, and must account for beginning or end of week crossovers. That is, if t_k is greater than 302,400 seconds, subtract 604,800 seconds from t_k. If t_k is less than -302,400 seconds, add 604,800 seconds to t_k.

IS-GPS-200 Table 20-IV

Elements of Coordinate Systems (3 of 3)

Table 20-IV. Elements of Coordinate Systems (sheet 2 of 2)

$E_k = \cos^{-1}\left\{\frac{e + \cos \nu_k}{1 + e \cos \nu_k}\right\}$		Eccentric Anomaly
$\Phi_k = \nu_k + \omega$		Argument of Latitude
$\delta u_k = c_{us}\sin 2\Phi_k + c_{uc}\cos 2\Phi_k$	Argument of Latitude Correction	Second Harmonic Perturbations
$\delta r_k = c_{rs}\sin 2\Phi_k + c_{rc}\cos 2\Phi_k$	Radius Correction	
$\delta i_k = c_{is}\sin 2\Phi_k + c_{ic}\cos 2\Phi_k$	Inclination Correction	
$u_k = \Phi_k + \delta u_k$		Corrected Argument of Latitude
$r_k = A(1 - e \cos E_k) + \delta r_k$		Corrected Radius
$i_k = i_0 + \delta i_k + (\mathrm{IDOT})\, t_k$		Corrected Inclination
$x_k' = r_k \cos u_k$ $y_k' = r_k \sin u_k$		Positions in orbital plane.
$\Omega_k = \Omega_0 + (\dot{\Omega} - \dot{\Omega}_e)\, t_k - \dot{\Omega}_e\, t_{oe}$		Corrected longitude of ascending node.
$x_k = x_k' \cos\Omega_k - y_k' \cos i_k \sin\Omega_k$ $y_k = x_k' \sin\Omega_k + y_k' \cos i_k \cos\Omega_k$ $z_k = y_k' \sin i_k$		Earth-fixed coordinates.